TATORT: EXCEL

A Forensic Guide to Data Cleaning · Excel for the Web

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T · POLIZEIABSPERRUNG · NICHT BETRETEN · CRIME SCENE · DO NOT CROSS · TATORT · POL

SONNTAG, 20:15 UHR

Since **29 November 1970**, Germany has settled in on Sunday nights for *Tatort* (“crime scene”), the longest-running crime drama on German-language TV.

- **1,311 episodes**, ten regional broadcasters, dozens of detective teams
- An episode list handled by many people over 55 years

Today the list itself is the crime scene. **You have 45 minutes.**

HOW TODAY WORKS

Pre-flight	Open your copy of the evidence	5 min
Sprint 1	Survey the scene	5 min
Sprint 2	Clear the scene	7 min
Sprint 3	Hidden characters	7 min
Sprint 4	Unmask the aliases	7 min
Sprint 5	Time of the crime	7 min
Debrief	What's left, and how to finish	3 min

Each sprint: **1-minute briefing → you work, timer on screen → checkpoint**. Fell behind? Every sprint has a **catch-up sheet** so you can jump straight back in.

MEET THE EVIDENCE

Sheet	What's in it
`Tatort_Evidence`	1,339 rows of episodes. Something is wrong with nearly every column.
`Formulas`	Every formula for today, ready to copy
`Known_Aliases`	Reference table: broadcaster alias → official code
`Data_Dictionary`	What each column means and what *clean* looks like
`Case_Log`	Your record of every change you make
`Catch_Up_2` – `Catch_Up_4`	The scene as it should look after each sprint

YOUR TURN

PRE-FLIGHT

⏱ 4 min | 🔗 Evidence link: add your short link here before the session

> ACTION
>
> Type the **evidence link** on screen into your laptop's browser. Signed in with your university account?

> ACTION
>
> **File → Create a Copy → Create a copy online.** Name it `Tatort_<your initials>`. Work only in **your** copy.

> ACTION
>
> Right-click the `Tatort_Evidence` tab → **Duplicate**. Rename the copy `Working`. Never edit `Tatort_Evidence` again.

Can't find a button? Type its name in Excel's **search box** at the top. AutoSave is always on: your duplicate tab *is* your undo button.

SPRINT 1

LOOK BEFORE YOU TOUCH

- Scroll to the **bottom** and the **far right**. Where does the data really end?
- **Data → Filter**, then open each column's drop-down. Every odd spelling is listed there.
- Click a cell and compare the **formula bar** with what the cell shows.
- Left-aligned "numbers" and "dates" are usually **text**.

> **CLUE**
>
> Write the clue down before you fix anything. Detectives who fix as they look lose track of what they found.

YOUR TURN

SPRINT 1: SURVEY THE SCENE

⏱ 3 min

ACTION

In `Working`: turn on **Data → Filter**. Open the drop-down of at least four columns.

ACTION

Write **three clues** on your Evidence Card. What's wrong, and in which column?

CHECKPOINT 1

Call out your clues. Did anyone find…

- three title rows above the header and junk columns on the right?
- the **same episode twice**?
- `wdr`, `W.D.R.` and `Westdeutscher Rundfunk`: **one broadcaster, many aliases**?
- dates that are left-aligned **text** in two different formats, and numbers like `25901`?
- titles in ALL CAPS, `n/a`, `unbekannt` and `NULL` instead of blanks?

Today we chase the first four. The rest are in the **On Your Own** guide.

SPRINT 2

CLEAR THE SCENE

- Row 1 **is** the header: delete the title rows above it
- Headers: short, unique, no stray spaces or ALL CAPS (match the `Data_Dictionary`)
- Delete columns you don’t need
- **Insert → Table**: new formulas fill the whole column automatically
- **Data → Remove Duplicates**, all columns ticked = only *exact* copies go

> **CAREFUL**
>
> Same title ≠ duplicate. *Die Abrechnung* is the title of three different episodes.

YOUR TURN

SPRINT 2: CLEAR THE SCENE

⏱ 5 min

ACTION

Select `A1:A3` → right-click → **Delete → Row**. Delete columns `Import_ID`, `Bearbeitet_von`, `Spalte1`.

ACTION

Fix three headers: `Folge` → `Folge`, `TITEL` → `Titel`, `Sender` → `Sender`.

ACTION

Click in the data → **Insert → Table** (✓ My table has headers). Then **Data → Remove Duplicates** → all columns → OK.

Write on your card: duplicates removed _____ · rows left _____

CHECKPOINT 2

DUPLICATES REMOVED

22

Got 0? You removed duplicates before deleting `Import_ID`.

ROWS LEFT

1,317

Episodes only go to 1,311, so six duplicates are still in there. See the On Your Own guide, Lead B.

TABLE?

✓

Click in the data: a **Table Design** tab appears.

Behind? Use the **`Catch_Up_2`** tab from here on.

SPRINT 3

HIDDEN CHARACTERS

Helper columns: never overwrite the evidence. Clean in a *new* column beside it, compare, keep both.

```
=TRIM(text)                          removes leading, trailing & double spaces
=TRIM(SUBSTITUTE(text, UNICHAR(160), " "))
                                     …and the invisible non-breaking space
=LEN(old)<>LEN(new)                  TRUE if cleaning changed something
```

Character 160 looks exactly like a space and comes in when data is copied from the web. `TRIM` alone ignores it.

YOUR TURN

SPRINT 3: HIDDEN CHARACTERS

⏱ 5 min | Behind? Start in `Catch_Up_2`

ACTION

New column **`Titel_Clean`**: copy its formula from the `Formulas` sheet.

ACTION

New column **`Titel_Changed`**: copy its formula too. Filter it for **`TRUE`**.

ACTION

How many? Copy the **Titles changed** counter from `Formulas` into an empty cell of `Case_Log` (column H). Write the number on your card.

CHECKPOINT 3

TITLES CHANGED

63

45 with extra spaces, 12 with a non-breaking space, and 6 near-duplicate rows that only differ by a trailing space.

WHY IT MATTERS

"LU " ≠ "LU"

To Excel, filters, lookups and Remove Duplicates, those are different values.

Behind? Use the `Catch_Up_3` tab from here on.

SPRINT 4

UNMASK THE ALIASES

`wdr`, `W.D.R.` and `Westdeutscher Rundfunk` are all **WDR**. Don’t Find & Replace 55 times: use a **reference table**.

```
=XLOOKUP(what, where to look, what to bring back, "if not found")

=XLOOKUP(TRIM(TEXTBEFORE([@Sender], "/", , , , [@Sender])),
         Known_Aliases!A:A, Known_Aliases!B:B, "CHECK")
```

- `TEXTBEFORE(…, "/")`: co-productions like `SWR/SRF` → lead broadcaster `SWR`
- `"CHECK"`: anything not on the list is flagged, not guessed

YOUR TURN

SPRINT 4: UNMASK THE ALIASES

⏱ 5 min | Behind? Start in `Catch_Up_3`

ACTION

Clear any filters. New column **`Sender_Code`**: copy its formula from `Formulas`.

ACTION

Open the `Sender_Code` filter drop-down. Any **CHECK**? How many broadcasters are left?

ACTION

Count WDR’s episodes: copy the **WDR episodes** counter into `Case_Log`, column H.

CHECKPOINT 4

CHECK

0

Every alias unmasked.

BROADCASTERS

15

Down from about 45 spellings in the filter list.

WDR EPISODES

238

The biggest of the 15. NDR is next with 198.

Behind? Use the `Catch_Up_4` tab from here on.

SPRINT 5

TIME OF THE CRIME

What you see	What it is
`1970-11-29` right-aligned	a real date
`25901`	a real date shown as a number (*General* format)
`29.11.1970` left-aligned	text, German day.month.year
`1970-11-29` left-aligned	text, ISO year-month-day

`DATE(year, month, day)` rebuilds text dates and **never guesses** your region. Then check: Tatort began on **29 Nov 1970**, so any earlier or future date is impossible.

YOUR TURN

SPRINT 5: TIME OF THE CRIME

⏱ 5 min | Behind? Start in `Catch_Up_4`

ACTION

New column **`Date_Clean`** from `Formulas`. Select the column → **Home → Number Format ▼ → Short Date**.

ACTION

New column **`Date_Check`** from `Formulas`. How many say **CHECK**? Use the **Impossible dates** counter.

ACTION

Clear the filter, then sort `Date_Clean` **oldest → newest**. What's the first real date?

CHECKPOINT 5

IMPOSSIBLE DATES

5

1907, 1958, 2030, 2091 and… 3799. Flag them; don’t guess.

FIRST REAL DATE

29 Nov 1970

Folge 1, *Taxi nach Leipzig*, the first episode.

TEXT DATES RESCUED

106

71 German + 35 ISO, plus 20 real dates that were only shown as numbers.

CASE NOT CLOSED (YET)

You cleaned the four biggest leads. The **On Your Own** guide walks through the rest, step by step:

- **A** · Yellow highlights
- **B** · Episode numbers & the last 6 duplicates
- **C** · Case numbers stored as text
- **D** · Writers: line breaks & `n/a`
- **E** · Directors & `NULL` notes
- **F** · ALL CAPS titles
- **G** · Investigators & guest stars
- **H** · PivotTable check
- **I** · File the report (CSV + Case Log)

GOOD HABITS

- **Secure the scene**: raw data is read-only
- **Look before you touch**, and write down what you see
- **Clean in helper columns**, compare, then swap in
- **Don’t trust what Excel shows you**: check with `LEN` and the formula bar
- **Don’t guess**: flag what you can’t verify
- **Document everything**: the Case Log is part of the evidence

YOUR TURN

BEFORE YOU GO

At the bottom of your Evidence Card:

> ACTION
>
> **One** technique from today you'll use on your own data, and *which* data.

> ACTION
>
> **One** thing that's still a mystery.

Materials, the On Your Own guide and the cheat sheet: see the link on your Evidence Card. Questions? Fragen?